Brisa+ Technical Report

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Urban Morphology and CFD Patch Sampling for Wind Simulation Across Five Informal Settlements in Rio de Janeiro

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Audience	Engineering reviewers; researchers reproducing or extending the pipeline; CFD team consuming the patch contract. *Not* a journal manuscript — that lives separately and cites this report.

For the **fastest engineering review path**, see `docs/onboarding/engineering_review.md`. For terminology, see **§0 Glossary** below; for reproducibility, see **§12 Reproducibility**; for failure modes and observability, see **§13 Failure modes**.

Executive Summary

This report documents the morphometric analysis and CFD sampling pipeline developed for five informal settlements in Rio de Janeiro: Vidigal, Rocinha, Rio das Pedras, Complexo do Alemão, and Maré. The work supports the Brisa+ project, which aims to quantify pedestrian-level wind conditions in informal urban fabric and link them to health-relevant outcomes (thermal comfort, pollutant dispersion, natural ventilation).

For an engineering reviewer, in three sentences. *What to use:* the 119-patch CFD sampling campaign (per-patch `buildings.gpkg` + `terrain.tif` + `patch_meta.json`) and the morphometric grid (10 m, 20+ indicators per cell) are stable and ready to consume — every output path is documented in §8 and reproducible from §12. *What not to trust yet:* anything labelled "annualised" or "wind-velocity coupled" in §11, because no real OpenFOAM results have returned from the cluster — the result-side pipeline is shipped and synthetic-validated end-to-end on all 5 sites, but the producer interface has not been

exercised against real data. *What's pending:* VDG-P07 ingestion (in flight at MIT ORCD), followed by incremental submission of the remaining 118 patches and annualisation against the measured wind roses.

At time of writing, the pipeline has produced:

- **98,435 building footprints** compiled from site-specific and city-wide cadastral data, with defensive geometry repair and extended context beyond each favela boundary.
- **82,314 grid cells at 10 m resolution** covering the five sites, each characterised by 20+ morphometric indicators including Sky View Factor, plan area density, 8-direction frontal area density, volumetric porosity, height variability, slope, aspect, and street orientation entropy.
- **119 CFD simulation patches** selected via stratified sampling across twelve morphometric strata (SVF × slope × λp) with maximin geographic spacing and SVF-priority weighting to oversample health-relevant low-sky- view conditions.
- **A complete CFD integration pipeline** (`src/cfd_integration/`, 63 passing tests) ready to ingest OpenFOAM results when the simulation campaign completes in the parallel CFD repository. The result-side analysis pipeline (`scripts/analyze_cfd_results.py`) is shipped and synthetic-validated end-to-end on all 5 sites; awaiting the first real return (VDG-P07).

The pipeline is fully reproducible from the committed scripts; all inputs are documented and all intermediate outputs preserved in the canonical repository layout. This document summarises the methodology, validates the sampling design, and specifies the interface between this repository and the CFD execution environment.

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0. Glossary / Nomenclature

Domain-specific terms used throughout this report. Each is defined once here, with units. Sections cite the term name; the definition lives only in this section.

Morphometric indicators (per 10 m grid cell)

Term	Symbol	Units	Definition
Sky View Factor	SVF	— (0–1)	Fraction of the upper hemisphere visible from a sample point at 1.5 m above ground. Cell value = mean over passageway samples whose centroid falls in the cell. Method: §4.2.
Plan area density (Building Coverage Ratio)	λp, BCR	— (0–1)	Sum of building footprint area in the cell ÷ cell area. Capped at 1.0 to prevent over-counting from overlapping footprints.
Frontal area density	λf	—	Projected vertical surface area per unit horizontal cell area, computed for 8 wind directions; stored as `lambda_f_{N,NE,…,NW}`, plus `lambda_f_mean` (mean across directions) and `lambda_f_max` (worst-case direction). Primary input to wind-canopy drag models.
Volumetric porosity	—	— (0–1)	`1 − (Σ building_volume_in_cell) / (cell_area × H_mean)`. Strongly anti-correlated with λp (r ≈ −0.95).
Height variability	σH	m	Standard deviation of building heights in the cell. NaN if < 2 buildings.
Mean building height	H_mean	m	Arithmetic mean of contributing building heights. Cell-level average — see §1 footnote on aggregation.
Max analysis-patch height	H_max_analysis	m	Tallest building inside a 100 m-diameter analysis patch (used for Blocken fetch check).
Slope, aspect	—	°, °	Cell-mean terrain slope and aspect from the merged DTM via numpy gradient. Aspect uses circular mean.
Street orientation entropy	—	— (0–1)	Normalised Shannon entropy of street-segment bearings (folded to [0, 180°)) within the cell. 0 = single direction, 1 = uniform.

Sky-view conventions

Term	Definition
Tregenza-145	Equal-area discretisation of the sky hemisphere into 145 patches (Tregenza, 1987). The IVF SVF engine ray-casts into the centroid of each patch from each sample point and reports the unobstructed fraction.
`svfForProcessing153`	UMEP's shadow-cast SVF processor with 153-patch sky discretisation (Lindberg & Holmer, 2010, used in SOLWEIG). Used as the independent benchmark in §4.2 / §10.3 cross-validation.
Height-matched	Both engines integrate at z = 1.5 m above ground, achieved in IVF via passageway sampling and in UMEP by lowering all building heights by 1.5 m before the shadow-cast.
Passageway aggregation	SVF cell value comes from samples on traversable ground (streets, alleys, courtyards) — never centroids inside building footprints, which would bias to 0.

CFD sampling

Term	Definition
Analysis patch	100 m-diameter circle (radius 50 m, area ≈ 7,854 m²) where pedestrian-level metrics are sampled.
CFD domain	250 m-radius circular domain enclosing the analysis patch; provides flow-development context per Blocken (2015).
12 strata	Cross-product of SVF (3 bins: < 0.15, 0.15–0.30, ≥ 0.30) × slope (2 bins: < 15°, ≥ 15°) × λp (2 bins: < 0.5, ≥ 0.5). Stratum IDs encoded `SVFn_SLPn_LPn`.
Maximin spacing	Greedy maximin geographic distance between patch centres, with an 80 m floor. Produces a spatially diverse sample within each stratum.
Blocken radius	Minimum CFD domain radius for adequate fetch development; the rule of thumb is 5 × H_max. The campaign uses a fixed 250 m radius; per-patch margin = 250 − 5·H_max_analysis.
Blocken margin	The actual numerical margin a patch has against the 5 × H_max constraint. For the 119-patch campaign: minimum 114 m (RDP-P15), maximum 215 m, median 180 m; 11/119 patches under 150 m, all at Rio das Pedras or Rocinha.
`blocken_ok`	Boolean field in `patch_meta.json` — true iff `5 × H_max_analysis ≤ cfd_domain_radius`. True for all 119 patches in the campaign.

Wind / atmosphere

Term	Definition
Wind rose	Per-site 8-direction frequency × mean-speed climatology, stored as `data/{site}/wind_rose.json`. Used post-hoc to weight the 8 directional CFD simulations into an annualised metric — *not* used for boundary conditions.
Calm fraction	Fraction of hourly observations with `|U| < 0.5 m/s` or direction = NaN. Excluded from direction binning but recorded explicitly.
Neutral log-law	Inflow profile assumed for all CFD runs: $u(z) = (u_*/\kappa) \ln((z+z_0)/z_0)$. Implies neutral atmospheric stability; deliberate methodological simplification (§10.1).
k-ω SST	Shear-Stress-Transport k-ω turbulence closure (Menter, 1994). Standard for ABL-scale RANS.
ACH	Air change rate per hour: `ACH = 3600 × ⟨|U|⟩ / L`, with L the canyon characteristic length.
TI	Turbulence intensity: `TI = √(2/3 · TKE) / U_ref`.
TKE	Turbulent kinetic energy.
Stagnation fraction	Fraction of patch sample points with `|U| < 0.5 m/s`.

Data sources

Term	Definition
DTM	Digital terrain model (bare-earth raster).
DSM	Digital surface model (DTM plus buildings/vegetation).
INMET BDMEP	*Banco de Dados Meteorológicos para Ensino e Pesquisa* — the Brazilian Met. Service's hourly archive. Source for 4 of 5 wind-rose stations (A652, A636, A621, A602).
ASOS	Automated Surface Observing System — Iowa State's archive of international airport METAR; source for SBGL Galeão (Maré).
METAR	Standardised aviation weather report format.
EPSG: 31983	SIRGAS 2000 / UTM Zone 23S — the project's canonical projected CRS.

1. Study Sites

Five favelas were selected to span the morphological typologies of Rio informal settlements:

Site	Area	Type	Buildings (extended)	10 m cells	Mean building height †
Vidigal	0.30 km²	hillside	4,600	3,169	6.3 m
Rocinha	0.80 km²	hillside	14,443	8,972	8.1 m
Rio das Pedras	0.70 km²	flatland	11,276	7,046	8.7 m
Complexo do Alemão	1.97 km²	mixed	28,783	19,708	5.3 m
Maré	4.34 km²	flatland	39,333	43,419	7.1 m
TOTAL	**8.11 km²**		**98,435**	**82,314**	—

Extended buildings = site footprints plus context buildings within a 300 m buffer (see Section 3.2). Typology assignment drives wind-regime analysis: hillside sites span 0–45° slopes; flatland sites cluster near 0°.

† Mean building height = mean of `H_mean` across all eligible 10 m grid cells per site (`outputs/{site}/morphometrics/grid/grid_metrics.gpkg`). This is a cell-area-weighted aggregation, not a per-building mean — short ancillary structures contribute less than the multi-storey blocks that fill more of the eligible cell area.

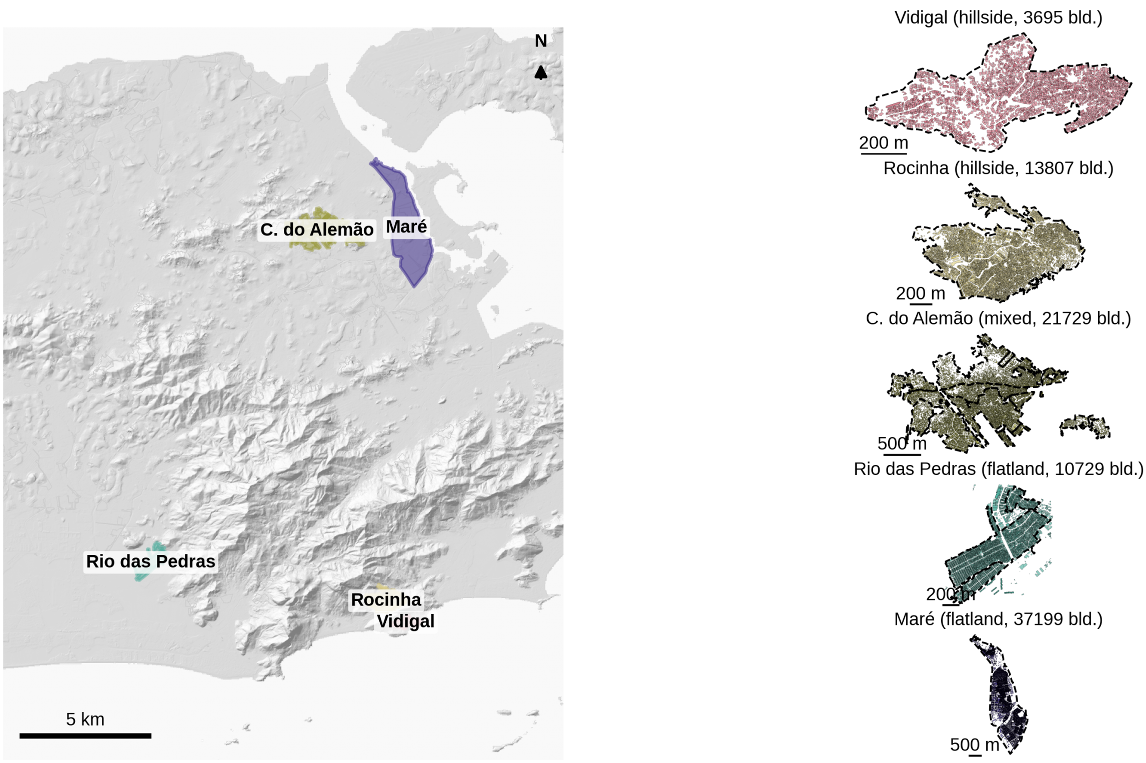

Figure 1. Overview of the five campaign sites on the Rio de Janeiro hillshade. The main panel shows the full metropolitan extent with all 1,074 registered favelas outlined in light grey; the five campaign sites are highlighted with typology-coloured fills. Right panels show each site's building footprints at a consistent visual style, with favela boundary and building count. Vidigal and

Rocinha occupy the steep south-zone hillslopes near Sugarloaf; Maré and Complexo do Alemão are in the flat north-zone bayside plain; Rio das Pedras is in the west-zone plain.

Cidade de Deus was initially considered but excluded from the CFD campaign due to a documented building-data defect (see `.claude/…/memory/project_cdd_data_bug.md`); its SVF values saturated at ≈ 0.90, indicating missing building footprints in the computational scene.

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2. Data Sources

2.1 Site-specific data

Each site has a `data/{site}/raw/` directory containing:

- **Building footprints** (.shp, 3D polygons) with attributes `altura` (height above base), `base` (base elevation), `topo` (top elevation). CRS: EPSG: 31983 (SIRGAS 2000 / UTM zone 23S).
- **Favela boundary** (.shp) from the 2019 Rio municipal cadastre.
- **Digital terrain model** (.tif) at 5 m resolution clipped to the site extent during data preparation (manually in GIS; not regenerated code-side per project memory).
- **Road network** (.shp) from the municipal `Logradouros` dataset, clipped and projected per site.

2.2 City-wide data (shared across sites)

Used to extend the building context and ensure continuous terrain for CFD domains that exceed individual favela boundaries.

- **`data/RJ/buildings_RJ_2019.shp`**: 2,362,806 building footprints for the entire municipality. Same schema as per-site files (subset extracted from this source).
- **`data/RJ/DTM_RJ.tif`**: 72 km × 37 km DTM at 5 m resolution covering all of Rio.
- **`data/RJ/Favelas_Limit_2019.shp`**: 1,074 registered favelas, used for the Figure 1 overview.

2.3 Wind forcing (INMET)

Annualised wind roses are required to weight CFD results across the eight cardinal directions. Each site has a `data/{site}/wind_rose.json` file whose schema is defined by `src.cfd_integration.schema.WindRose` — frequencies and mean speeds per direction, plus provenance metadata (station id, coordinates, time window, observation count, calm fraction, anemometer reference height, and a `quality_flag` ∈ {`measured`, `gap-filled`, `placeholder-prior`}).

Current status (April 2026). All five `wind_rose.json` files now carry measured hourly observations across the full 2015–2024 window and are tagged `quality_flag: "measured"`. The four hillside / inland sites use INMET BDMEP records; Maré uses the Iowa State ASOS METAR archive of SBGL Galeão. Sample sizes range from n = 64,088 (A636 Jacarepaguá, which only began reporting in August 2017) to n = 89,439 (SBGL). Per-site directional frequencies, mean speeds, and calm fractions are shown in **Figure S5** (`docs/technical_report/figures/figS5_wind_roses.png`):

Site (station)	n	Window	Calm	Prevailing dir	Notes
Vidigal / Rocinha (A652 Forte de Copacabana)	85,103	2015–2024	1.4 %	E (30 %), W (23 %)	Coastal cliff: bimodal sea-breeze and reverse-channel along the Dois Irmãos saddle.
Rio das Pedras (A636 Jacarepaguá)	64,088	2017–2024	46.3 %	SW (26 %), W (20 %)	Sheltered basin, weak winds ($\bar{U} \approx 1.3$ m/s); high calm fraction reflects the Jacarepaguá lowland regime, not measurement gaps.
Complexo do Alemão (A621 Vila Militar)	86,019	2015–2024	33.1 %	N (25 %), SW (19 %)	North-zone interior; bay sea-breeze + post-frontal SW pattern.
Maré (SBGL Galeão METAR)	89,439	2015–2024	3.7 %	SE (26 %), E (19 %)	Bay regime; continuous METAR coverage gives the largest sample.

Recommended stations (verified April 2026 against the INMET catalogue, daily-graph URLs, and the published TMY paper for A652):

Site	Station	Code	Coords (lat, lon)	Class	Notes
Vidigal	Forte de Copacabana	A652	−22.988, −43.190	coastal	Nearest unobstructed coastal reference (~5 km E). CFD captures the Dois Irmãos lee-side locally; inflow rose stays at A652.
Rocinha	Forte de Copacabana	A652	−22.988, −43.190	coastal	Provides the unobstructed SE→NE driver. Valley channelling is resolved by the CFD itself.
Rio das Pedras	Jacarepaguá	A636	−22.99, −43.37	plain	Colocated with the Jacarepaguá lowland.
Complexo do Alemão	Vila Militar	A621	−22.86, −43.41	urban interior	Closest north-zone station (~8 km W). Corrects the earlier placeholder recommendation of A602 Marambaia, which is geographically mismatched (southwest coast, not north zone).
Maré	SBGL Galeão METAR (preferred); A652 (INMET fallback)	— / A652	−22.81, −43.25 / −22.988, −43.190	bayside	Galeão airport METAR via the Iowa State ASOS archive is the best match for the bay regime; ingested via `build_wind_rose.from_iowa_asos_csv`. A652 is the INMET fallback.

The earlier placeholder attributed A652 to "Alto da Boa Vista" — that name belongs to the municipal Alerta Rio network, not INMET. A652 is Forte de Copacabana. The station table is corrected here.

Data acquisition. INMET publishes one nationwide yearly ZIP (~100 MB) at `https://portal.inmet.gov.br/uploads/dadoshistoricos/{YEAR}.zip`, live and unauthenticated (browser User-Agent required). Each ZIP contains one CSV per automatic station for the year. The ingestion pipeline is documented in `src/cfd_integration/README.md` and implemented in `scripts/build_wind_rose.py from_inmet_csv`.

BDMEP CSV format: `sep=;`, `decimal=,`, latin-1 encoding, 8-row metadata header, missing values encoded as −9999, anemometer at z = 10 m. Calm periods (|U| < 0.5 m/s or direction = NaN) are excluded from direction binning but recorded in `calm_fraction` so they are not hidden.

Neutral-stability assumption. The k-ω SST inlet uses a log-law profile that implicitly assumes a neutral atmospheric boundary layer. For Rio, daytime unstable convection and evening stable inversions bias the stagnation metric. This limitation is accepted for the screening campaign and documented in §10. A future campaign could stratify the rose by stability class (e.g., day vs night, or Pasquill-Gifford class).

2.4 Coordinate reference system

All vector and raster data are in **EPSG:31983 (SIRGAS 2000 / UTM zone 23S)**, verified on load. Mixed-CRS inputs are reprojected automatically in the pipeline.

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3. Data Preparation

3.1 Building footprint processing

Raw footprints undergo three processing steps before use:

1. **Topology validation.** Shapely's `is_valid` flag is checked on all geometries. Rocinha was found to have 10 self-intersecting polygons (0.07 %) out of 13,807; all other sites had zero invalid geometries. All invalids are self-intersections (common artefact of municipal GIS digitisation) and are repaired by `buffer(0)`, which corrects the geometry with sub-square-metre area change. The repair is applied defensively in both the extended-context builder and the morphometric pipeline before any spatial operation. Full audit: `outputs/comparative/audits/rocinha_topology_audit.md`.

2. **Height validation.** `altura` values are checked for negatives (none) and unrealistic extremes (one building > 50 m — a legitimate high-rise in context data near a favela).

3. **CRS enforcement.** If source CRS differs from EPSG:31983, the dataset is reprojected.

3.2 Extended building context

Problem. Each CFD simulation domain is a circular region of 250 m radius around the analysis patch. When the patch is near the favela perimeter, the domain extends beyond available building data, leaving the outer annulus without context. A naïve sampling restricted to patches fully within the favela excludes 55 % of candidate cells at Vidigal.

Solution. The city-wide RJ building dataset is clipped to a 300 m buffer around each favela boundary and merged with the site-specific dataset, preferring site data where footprints overlap (deduplication via spatial join on the boundary polygon).

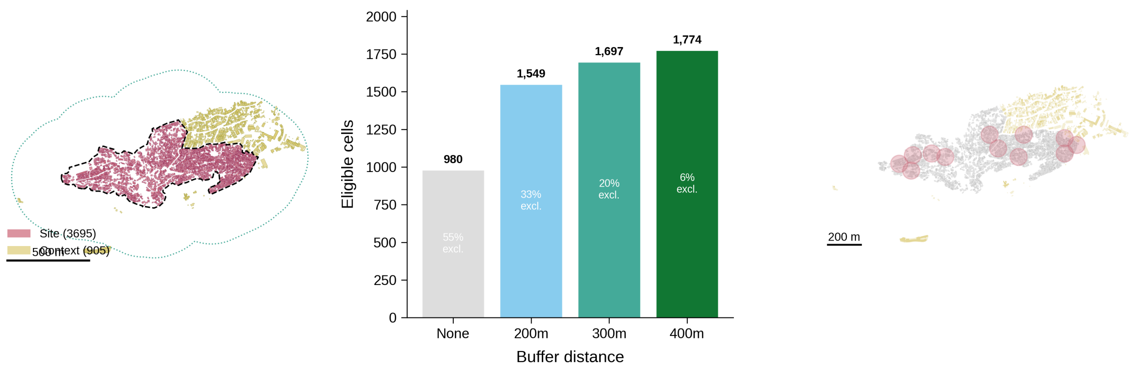

Figure S2. Context extension for Vidigal. Left: site buildings (blue) within the boundary and context buildings (orange) from city-wide data within the 300 m buffer. Middle: candidate-pool sensitivity to buffer distance — 0 m yields 980 eligible cells (31 %); 300 m yields 1,697 (54 %); 400 m only adds 77 more (diminishing returns). Right: pilot patch locations on the extended footprint — without the extension, patches cluster on the interior 600 m × 250 m; with it, they span the full 1,040 m × 310 m.

Extended buildings per site:

Site	Site buildings	Context buildings	Extended total
Vidigal	3,695	905	4,600
Rocinha	13,807	636	14,443
Rio das Pedras	10,729	547	11,276
Complexo do Alemão	21,729	7,054	28,783
Maré	37,199	2,134	39,333

Implementation: `scripts/build_extended_context.py`. The 300 m buffer was selected as the inflection point of a sensitivity scan (200 m / 300 m / 400 m); 300 m reduces domain-extent exclusion to ≤ 20 % at all sites while adding modest file size.

3.3 Extended terrain

The site-specific DTM is merged with the city-wide RJ DTM (both 5 m resolution) over the same 300 m buffer. The site DTM takes priority in overlap regions (`rasterio.merge.merge` with `method="first"`). A post-merge sanitisation step replaces any residual float32-max sentinel values (historical nodata artefact) with a clean nodata = −9999 and declares it in the output profile. This was found to be necessary after the first run produced downstream warnings about values of order 3.4×10^{38} in the merged raster.

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4. Morphometric Grid

4.1 Grid construction

A regular 10 m grid is generated covering each extended-buildings extent, then clipped to the favela boundary so that only cells whose centroid falls within the settlement are retained. Cells range from 3,169 at Vidigal to 43,419 at Maré, with total 82,314 cells across the five sites.

4.2 Indicators

Per 10 m cell, the pipeline computes:

Sky View Factor (SVF). The fraction of hemispheric sky visible from pedestrian height. Computed at 1.5 m above the ground surface via Tregenza 145-patch ray-casting on a sample grid of passageway points; the per-cell value is the mean of all sample points whose centroid falls within the cell. An `svf_count` column records the number of contributing sample points per cell (mean per site ranges from 1.9 (Maré) to 12.9 (Vidigal); pooled mean 3.5; minimum 1). Cells with no contributing samples receive NaN. The cross-site spread reflects passageway density: hillside fabrics with narrow, frequent corridors yield more samples per cell than flatland blocks where samples concentrate in a few wide streets.

Critically, SVF is **aggregated from passageway samples, not computed at grid-cell centroids**. Centroid computation would bias SVF toward 0 for cells whose centroid falls inside a building; passageway aggregation avoids this artefact. A diagnostic verification confirmed no spurious SVF = 0 spikes (see `src/morphometry/grid.py::_aggregate_svf_to_grid` and `docs/technical_report/validation.md`).

Cross-validation against UMEP. The Tregenza-145 ray-cast SVF was benchmarked against UMEP's shadow-casting SVF processor (`svfForProcessing153`, Lindberg & Holmer 2010, used in SOLWEIG) on a 1 m digital surface model rasterised from the same building footprints and DTM. UMEP averages over 153 hemispherical patches via shadow-casting on the DSM rather than ray-casting on a 3D mesh; this is an algorithmically distinct independent reference. To make the two engines height-comparable, we lower every building height by 1.5 m before running UMEP — equivalent to lifting the integration plane to pedestrian height — and mask rooftop pixels (UMEP otherwise includes them, IVF does not because passageway samples never fall on roofs). Aggregating UMEP at the same 10 m grid across all five sites:

Site	n	r^2	slope	RMSE	bias (UMEP − IVF)
Vidigal	2,510	0.68	0.96	0.12	+0.01
Rocinha	5,646	0.81	1.01	0.14	+0.09
Rio das Pedras	1,503	0.93	1.25	0.11	+0.08
Complexo do Alemão	4,838	0.76	0.92	0.17	+0.14
Maré	9,516	0.94	0.97	0.12	+0.09

Across all five sites the engines agree on the same physical quantity: four of five regression slopes (Vidigal 0.96, Rocinha 1.01, Complexo do Alemão 0.92, Maré 0.97) fall within ±8 % of unity. Rio das Pedras is the slope outlier at 1.25 — the site has the smallest valid grid (n = 1,503 after eligibility filtering), and its eligible cells are concentrated in the 0.3–0.5 SVF range where the 153-patch shadow-casting integration picks up partial obstruction that the 145-patch ray-cast smooths through. This is a sensitivity to grid coverage at smaller settlement extents, not a discrepancy in how the two engines compute SVF.

The r^2 range (0.68–0.94) tracks site relief: low-relief, spatially coherent fabrics (Maré 0.94, Rio das Pedras 0.93) yield the strongest per-cell agreement because adjacent cells share local terrain and similar average obstruction. High-relief hillside settlements (Vidigal 0.68, Complexo do Alemão 0.76, Rocinha 0.81) produce more cell-to-cell variation between integrators because each cell sits in a distinct local viewshed.

All five biases are positive (UMEP integrates over more sky than IVF), ranging from +0.01 (Vidigal) to +0.14 (Complexo do Alemão). The direction is consistent with the +1.5 m raised-observer transform applied to building heights before running UMEP: structures shorter than 1.5 m get partially or fully zeroed in IVF's input, opening additional sky in the UMEP comparison. Sites with substantial single- storey residential coverage (Complexo do Alemão, Maré, Rocinha, Rio das Pedras) sit in the +0.08 to +0.14 range; Vidigal's +0.01 reflects a footprint dominated by taller multi-storey blocks where the height shift has limited effect. The Vidigal bias collapse from −0.05 (z = 0, prior comparison) to +0.01 (z = 1.5) confirms the systematic offset is the sampling-height difference rather than the algorithmic distinction between ray-casting on a 3D mesh and shadow-casting on a DSM. See `scripts/validate_svf_against_umep.py` and `outputs/{site}/morphometrics/svf/umep_validation/`.

Plan area density (λp, BCR). Building footprint area ÷ cell area. Capped at 1.0 to prevent over-counting from overlapping footprints in dense informal fabric.

Frontal area density (λf). Projected vertical surface area per unit horizontal cell area, computed for eight wind directions (N, NE, E, SE, S, SW, W, NW); stored as `lambda_f_{direction}` plus `lambda_f_mean` and `lambda_f_max`. Directional λf is the primary morphological input to wind-canopy drag models.

Volumetric porosity. `1 − (Σ building_volume_in_cell) / (cell_area × H_mean)`. Characterises the void fraction in the urban canopy; strongly anti-correlated with λp (r ≈ −0.95 across sites, see Section 5.3).

Height variability (σH). Standard deviation of building heights in the cell. Cells with < 2 buildings receive NaN.

Mean height (H_mean). Arithmetic mean of contributing building heights.

Street orientation entropy. Shannon entropy of street-segment bearings clipped to the cell (bearings folded to [0, 180°) because streets are bidirectional), normalised to [0, 1]. Measures how ordered the street grid is: 0 = single dominant direction, 1 = uniform distribution across 36 bins.

Terrain slope and aspect. Derived from the merged DTM via numpy gradient, sampled within each cell and averaged. Aspect uses a circular mean to avoid discontinuity at 0°/360°.

Implementation: `src/morphometry/grid.py`. The computation is deterministic and takes ~30–80 s per site (longest is Complexo do Alemão at ~85 s) on standard hardware.

4.3 Distributions across sites

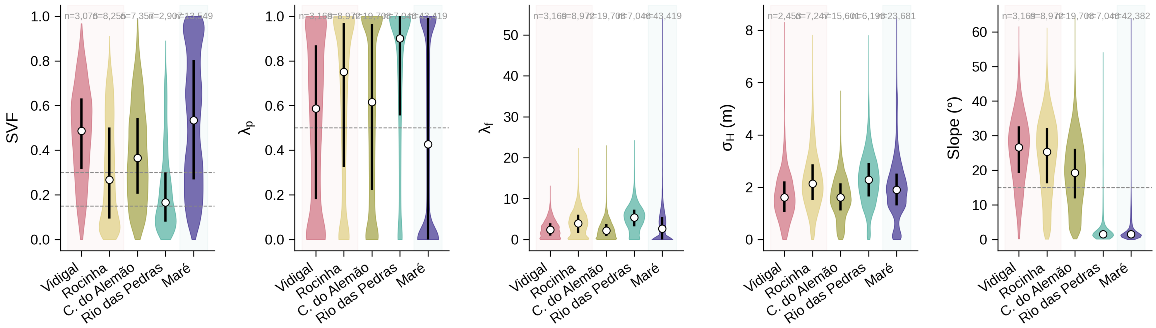

Figure 2. Distributions of the five primary indicators (SVF, λp, λf, σH, slope) across all five sites. Violins show the full distribution per site; white dots are medians and dark bars span the IQR. Dashed horizontal lines mark stratum boundaries (SVF at 0.15 and 0.30; λp at 0.5; slope at 15°). Background shading groups hillside (left, warm) and flatland (right, cool) sites.

Key patterns:

- **Slope clearly separates typologies.** Hillside sites (Vidigal, Rocinha) have most cells at 15–35°; flatland sites (Rio das Pedras, Maré) concentrate below 5°; Complexo do Alemão spans both regimes.
- **λp medians span 0.43 (Maré) to 0.90 (Rio das Pedras)** — denser than is typical of formal urban fabric. Maré, with its planned-housing block interiors, sits at the low end; Rio das Pedras, the densest informal infill, at the high end.
- **SVF distributions differ substantially.** Rocinha's dense canopy produces a strongly left-skewed distribution (median 0.27); Maré's more open layout has a right-skewed distribution (median 0.53).
- **σH is small everywhere** (median 1.6–2.3 m, span across all five sites), reflecting the typical 2–3 storey construction.

4.4 Spatial visualisation

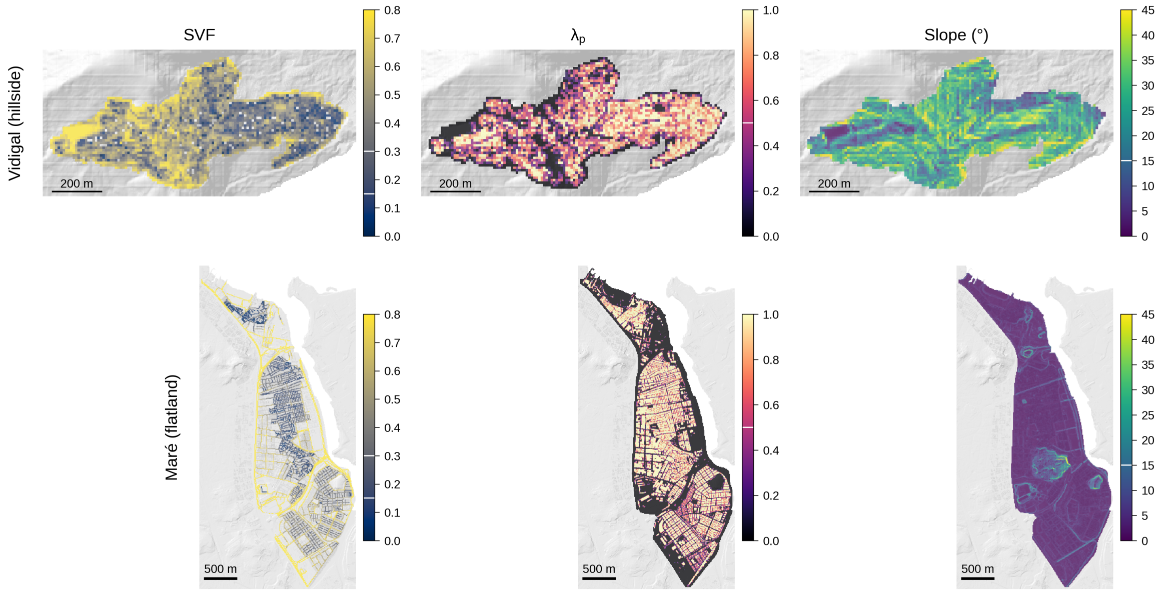

Figure 5. Spatial maps of SVF, λp, and slope for Vidigal (hillside) and Maré (flatland), with hillshade basemap and building outlines. Colour bars show indicator ranges; white contour lines on colourbars mark the stratification thresholds.

Notable observations:

- Vidigal's SVF map shows clear low-SVF corridors along the hillslope ridgelines where canopy density is highest (dark cyan in upper-left panel).
- Maré's λp is near-uniform at 0.7–1.0 across most of the settlement, reflecting the tight block structure of the original housing project layout.
- Slope maps confirm Vidigal's topography (green-yellow, 20–40°) vs Maré's flatness (purple, 0–3° except at engineered berms near the bay).

4.5 Indicator correlations

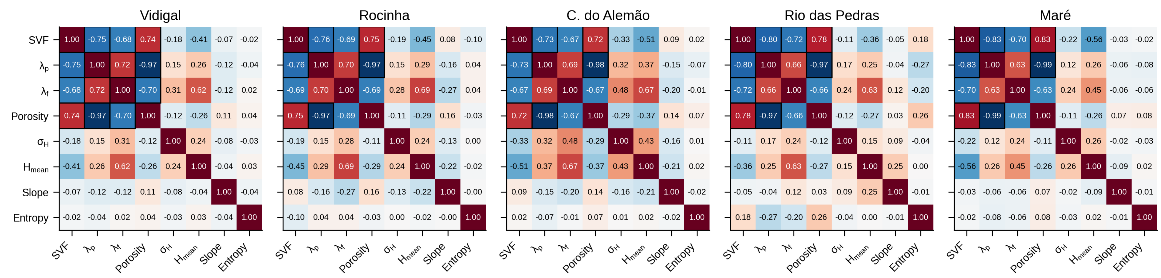

Figure S1. Pearson correlation matrices for the eight primary indicators at each site. Values are annotated in each cell; pairs with |r| > 0.7 are framed in black. Colour scale is fixed across all panels so patterns are directly comparable.

- **λp ⇄ porosity** is the strongest correlation everywhere (r ≈ −0.95), which motivates dropping porosity from the manuscript's distribution figure (Figure 2) to avoid redundancy.
- **SVF ⇄ λp** correlation is strong but not perfect (r ≈ −0.73 to −0.83 across the five sites): variation in building *height* and arrangement breaks the λp↔SVF link at fine scales. This is the structural-coupling finding developed in Section 5.
- **σH correlates weakly with H_mean** (per-site r ≈ 0.15–0.43, pooled ≈ 0.25). Earlier prose claimed r ≈ 0.5; the current cross-site number is lower because the bimodal "regular block + tall outliers" pattern saturates in dense favela fabric.
- **Slope is nearly uncorrelated with morphology**, confirming that terrain and urban form are independent stratification axes — a necessary condition for the 12-stratum sampling scheme.

4.6 Resolution sensitivity

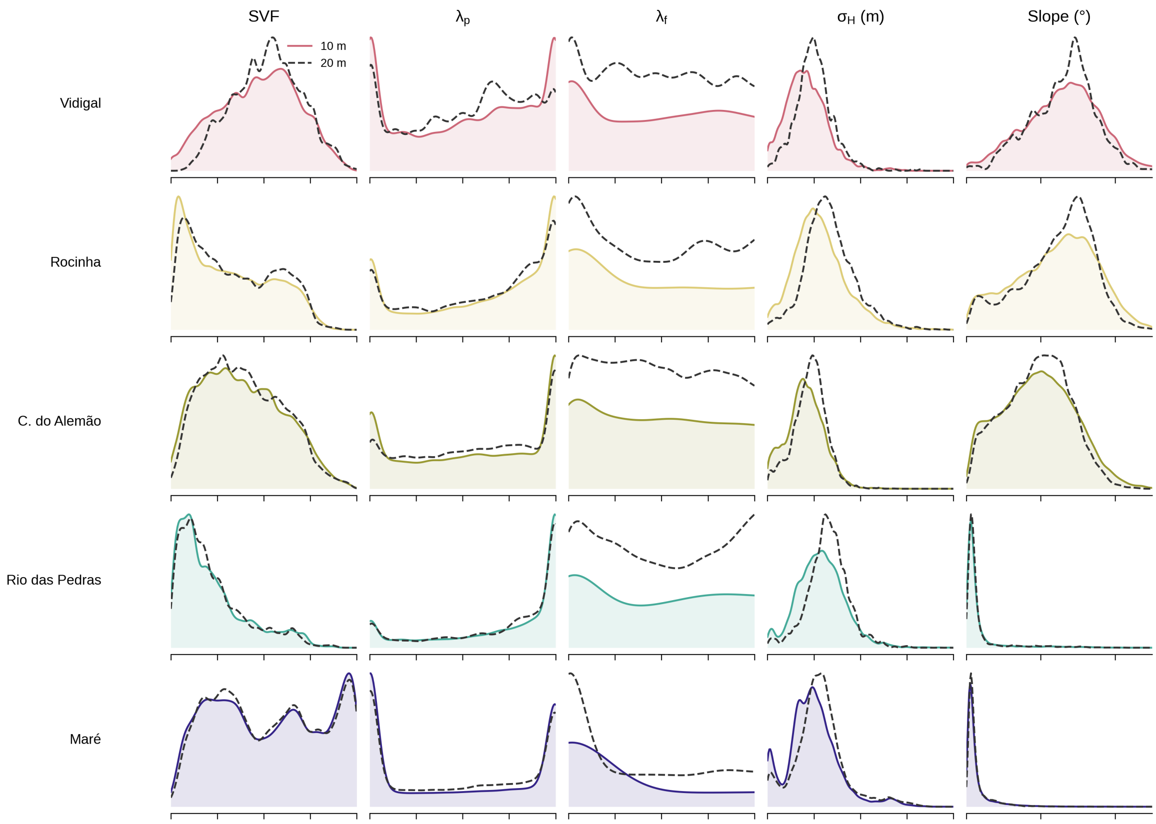

Figure S3. 10 m (solid coloured curves) vs 20 m (dashed black curves) morphometric distributions across all five sites and five indicators. The 10 m grid captures multimodal structure in λp and λf that 20 m aggregation smooths away — particularly visible in Complexo do Alemão and Rio das Pedras where the 20 m curves flatten features that clearly exist in the 10 m data.

This analysis justifies the 10 m resolution as the production grid:

- The key finding is preserved — distribution shape is similar — confirming that 10 m is not noise.
- But fine-scale multimodality relevant to CFD boundary conditions is only resolved at 10 m.
- Computational cost at 10 m is manageable (< 85 s per site).

Supplementary variants (Vidigal-only upscaled-vs-native scatter, and side-by-side difference maps) are archived in `outputs/paper_figures/exports/_variants/` and regenerable with `figS3_resolution_sensitivity.py --variants`.

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5. Cross-Site Morphology

5.1 SVF–λp structural coupling

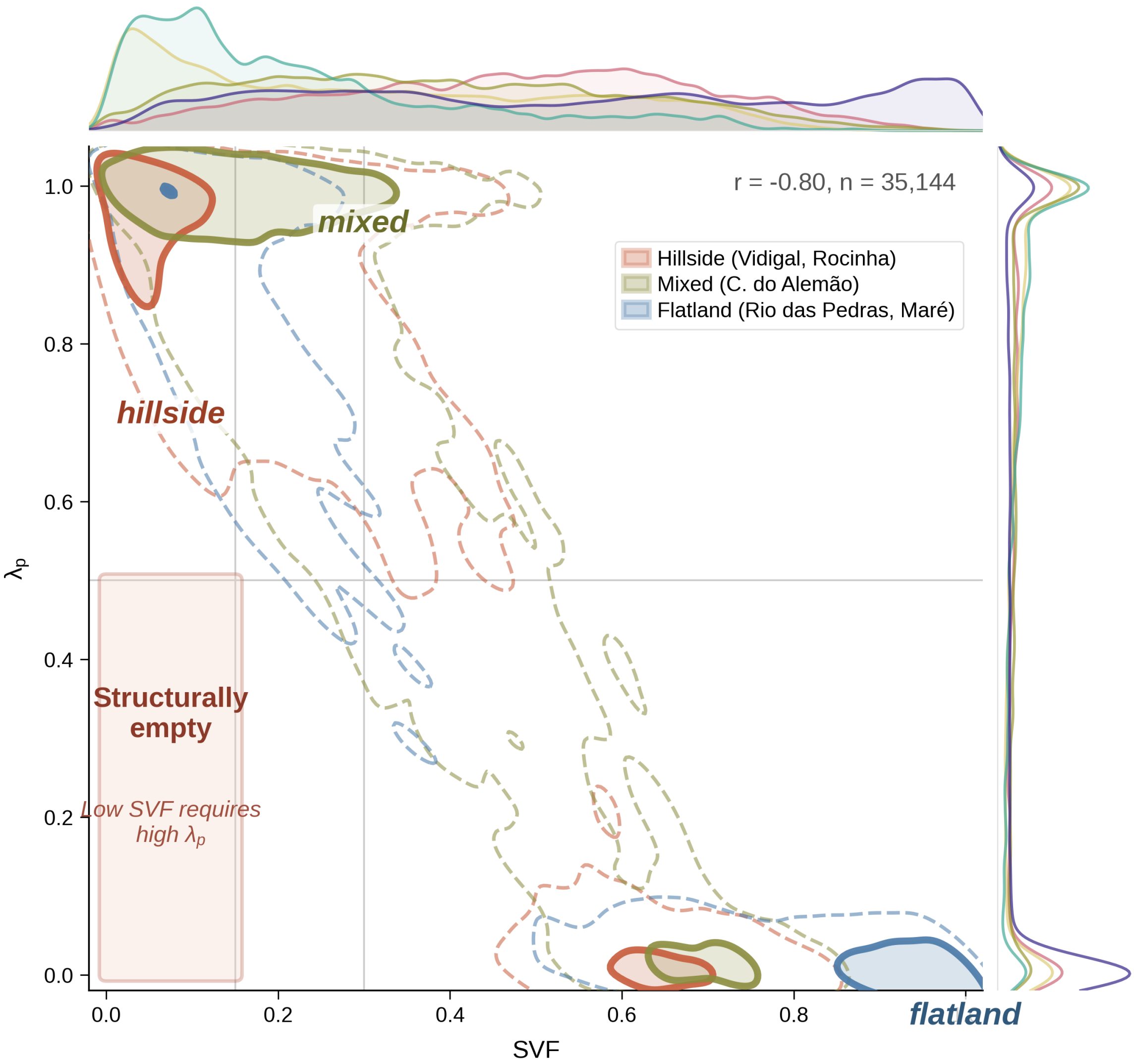

Figure 3. Central panel: 2D kernel density estimation of cells in (SVF, λp) space, grouped by typology. Solid contours enclose 50 % of each typology's cells; dashed contours enclose 90 %. Terracotta = hillside (Vidigal, Rocinha); olive-gold = mixed (Complexo do Alemão); steel blue = flatland (Rio das Pedras, Maré). Marginals show per-site SVF and λp distributions. The salmon-shaded region in the lower-left (SVF < 0.15, λp < 0.50) is **structurally empty** — only 102 cells in 35,144 (0.3 %) — because low sky view requires high plan density in realistic urban fabric. The strong negative correlation (r = −0.80 across all pooled cells) reflects this coupling. The three typologies occupy distinct but partially overlapping regions of the feature space: hillside cells cluster at higher λp for a given SVF, consistent with taller canopies; flatland cells spread to lower λp.

This figure is the central conceptual claim for the manuscript: the morphology of informal settlements occupies a constrained subset of the (SVF, λp) plane, and stratified CFD sampling must respect this constraint (empty strata are not allocation failures but physical impossibilities).

5.2 Typology summary statistics

Metric	Hillside (VDG+ROC)	Mixed (CDA)	Flatland (RdP+MAR)
Slope, median	25.7°	19.3°	1.5°
SVF, median	0.34	0.37	0.43
λp, median	0.71	0.62	0.54
λf_mean, median	3.43	2.20	3.12
σH, median	2.0 m	1.6 m	2.0 m

Source: medians computed from the concatenation of `outputs/{site}/morphometrics/grid/grid_metrics.gpkg` across the 12-strata eligible cells per site. Note the *Mixed* row reflects only Complexo do Alemão (which spans both flat and steep terrain — its slope median sits between the typology extremes); aggregating it with either Hillside or Flatland would compress that signal.

Two observations matter for the CFD design. First, **λp is uniformly high (0.54–0.71)** across all three typologies — informal urban form is denser than formal Rio fabric (typical λp in Copacabana ≈ 0.40), so the sampling allocation correctly prioritises high-λp strata. Second, **hillside and flatland sites carry comparable λf_mean (3.43 vs 3.12)**; the Mixed typology dips because Complexo do Alemão has the lowest building heights of the campaign (mean H_mean = 5.3 m). Wind-canopy drag will therefore not separate cleanly along the hillside↔flatland axis; SVF and slope are the dominant discriminators that CFD will quantify.

---

6. CFD Patch Sampling

6.1 Sampling design

Each CFD simulation corresponds to one **analysis patch** of interest — a 100 m-diameter circle (radius 50 m, area ≈ 7 854 m²) — embedded in a 250 m-radius circular domain that provides flow-development context (per Blocken 2015 and COST Action 732). The circular patch shape matches the cylindrical symmetry of the CFD domain, avoids corner artefacts in morphometric averaging, and produces isotropic coverage independent of building-grid orientation. To characterise the morphological diversity of the five sites with a finite simulation budget, patches are allocated across **twelve stratification bins**:

- SVF: 3 bins (SVF < 0.15, 0.15 ≤ SVF < 0.30, SVF ≥ 0.30)
- Slope: 2 bins (< 15°, ≥ 15°)

- λp: 2 bins (< 0.5, ≥ 0.5)

giving 3 × 2 × 2 = 12 strata, named e.g. `SVF1_SLP2_LP2` for SVF < 0.15 + slope ≥ 15° + λp ≥ 0.5.

6.2 Eligibility filter

A cell is an eligible patch centre if:

1. The 250 m CFD domain is fully covered by available building data (the extended buildings from Section 3.2). Measured as the fraction of the circular domain that intersects the convex hull of the extended footprints; threshold 0.7.
2. Within the 100 m-diameter circular analysis patch, building footprint coverage (∑ intersection area ÷ $\pi \cdot 50^2 \approx 7\,854$ m²) ≥ 0.5. This also implicitly rejects patches clipped by the site boundary, where the missing area fails the coverage threshold.
3. All three stratification indicators are non-null.

Filter outcomes per site:

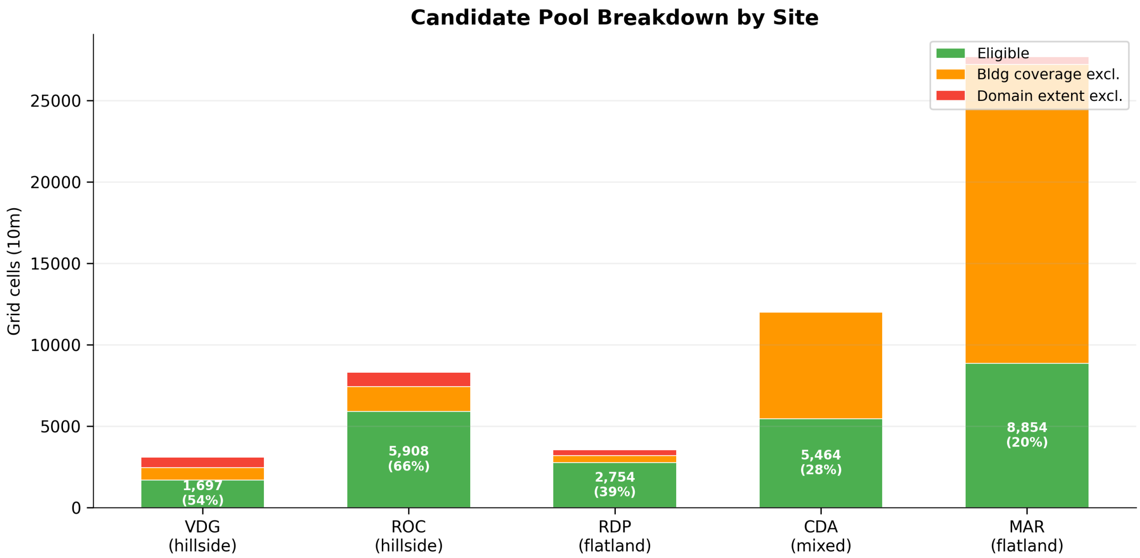

Figure (candidate pool). Stacked bars showing eligible cells (green) vs excluded cells per filter. Two exclusion sources: domain-coverage failure (red, dominated by edge cells in Vidigal and Rio das Pedras) and low-building-coverage failure (orange, dominant in Maré because of large residential block interiors without continuous footprint).

With the 300 m building-context extension, the domain-coverage exclusion drops from 55 % (no extension) to 0–20 % depending on settlement shape. The remaining exclusions are legitimate morphology: Maré's 20 % overall eligibility rate reflects its lower footprint coverage per 100 m patch, not a sampling artefact.

6.3 Selection algorithm

For each site, patches are chosen in two phases:

Phase 1 — Pilot (12–15 patches per site). Within each non-empty stratum, at least one patch is allocated; an additional bonus is given to the three most populated strata. Patches are chosen by **greedy maximin geographic distance** starting from the two most distant eligible cells; subsequent picks maximise minimum distance to already- selected patches (and to patches of other strata selected earlier, so that the cross-stratum spacing constraint is respected). Minimum spacing: 80 m between patch centres. With 100 m-diameter patches, two patches at the minimum centre-to-centre distance overlap in a lens of ≈ 817 m^2 (≈ 10.4 % of each patch area); this residual overlap is accepted because the patches are evaluated independently in separate CFD simulations and no joint statistics are computed over overlapping footprints.

Phase 2 — Campaign (incremental to 22–25 patches per site). Pilot patches are held fixed and treated as immovable seeds; additional patches are chosen with **SVF-priority weighting** to oversample the health-relevant low-SVF strata:

- SVF1 (< 0.15) → ×2.0 multiplier
- SVF2 (0.15–0.30) → ×1.0
- SVF3 (≥ 0.30) → ×0.8

The effective allocation is proportional to the weighted eligible count per stratum, subject to per-stratum minimums (≥ 1 patch if any cells exist) and the 80 m spacing constraint against the union of pilot and newly-selected patches.

Implementations: `scripts/run_pilot_sampling.py` (Phase 1), `scripts/run_campaign_sampling.py` (Phase 2), `scripts/build_extended_context.py` (upstream).

6.4 Allocation results

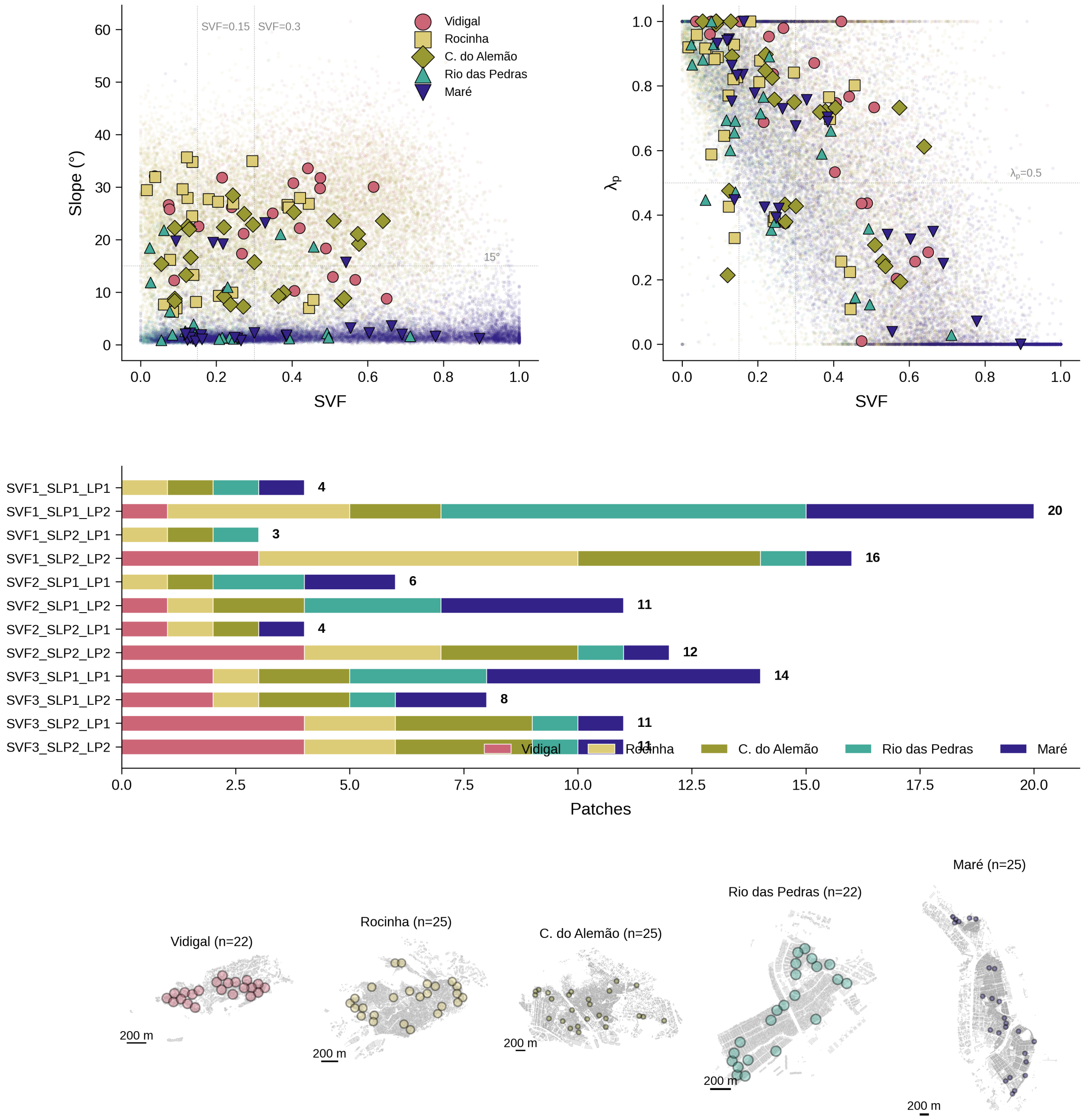

Figure 4. (a–b) Feature-space coverage of the 119 selected patches over the full grid-cell cloud for (SVF, slope) and (SVF, λp) axes. Dotted lines mark stratum boundaries. (c) Horizontal bar chart of patches per stratum, with segments coloured by site; numeric labels give the stratum total. (d) Per-site maps showing analysis-patch locations as coloured 100 m-diameter circles over grey building footprints. All five sites shown at consistent visual scale with 200 m scale bars.

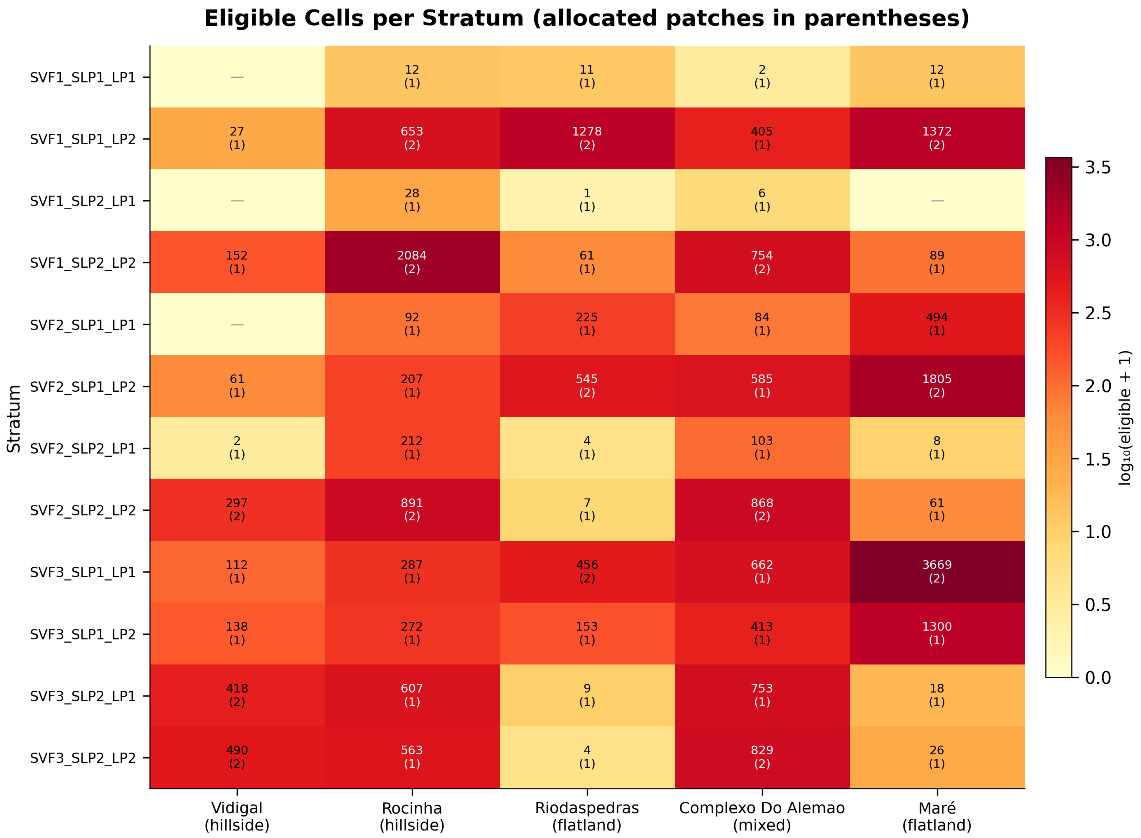

Figure (strata heatmap). Matrix of eligible cell counts ($\log_{10}$ scale) and allocated patches per stratum per site. Cells with no eligible candidates are marked with an em-dash. This makes the partial-empty structure explicit: three strata are empty at one or more hillside sites (SVF1_SLP1_LP1, SVF1_SLP2_LP1, SVF2_SLP1_LP1) because Vidigal's narrow hillside settlement has no flat cells with sparse footprints.

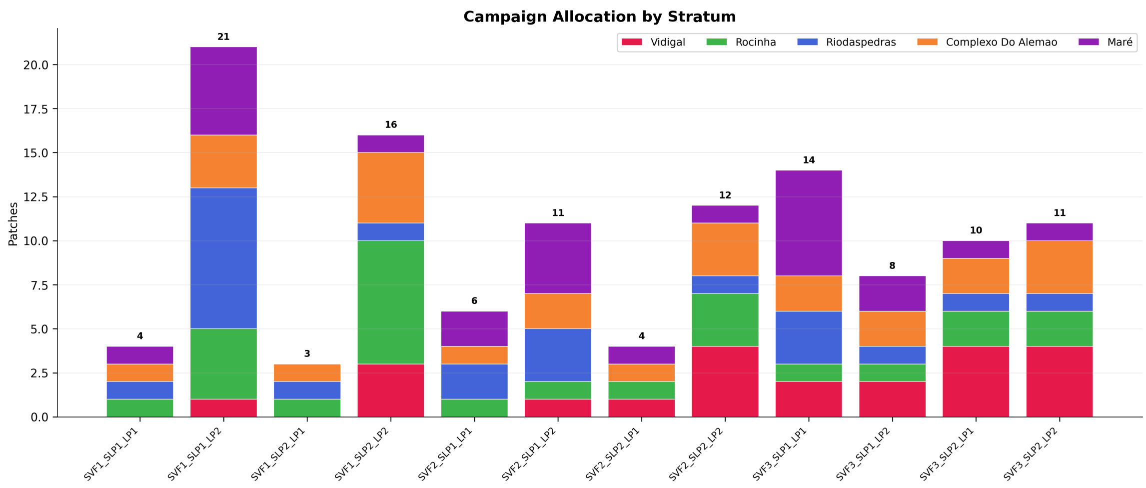

Figure (campaign allocation). Total patches per stratum in the final 119-patch campaign, segments coloured by site. SVF1_SLP1_LP2 (low sky view, flat, dense) has the highest patch count (20) after SVF-priority weighting — this is the stratum most relevant to health outcomes at the flatland sites and was intentionally oversampled.

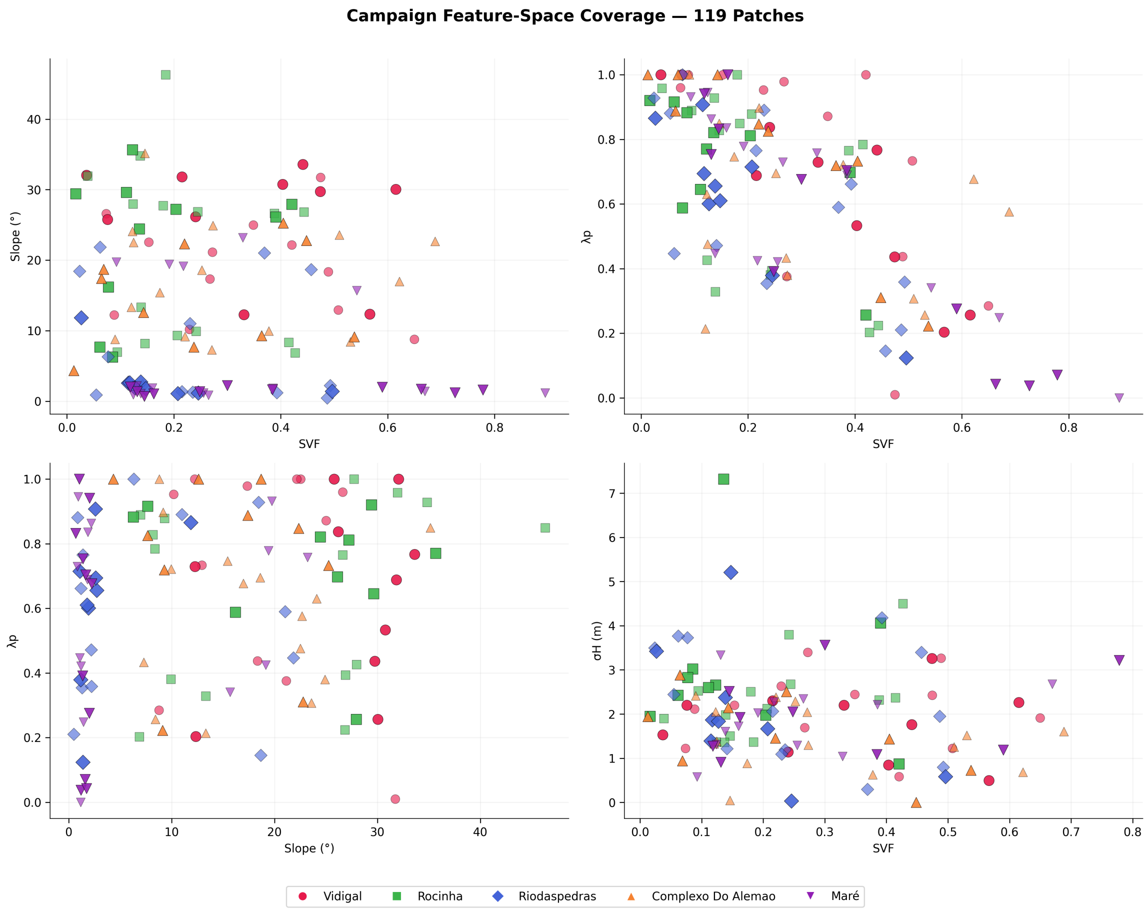

Figure (cross-site feature space). All 119 patches overlaid on all five sites' grid-cell clouds across four feature-space projections (SVF vs slope, SVF vs λp, slope vs λp, SVF vs σH). Site colours match throughout. Verifies that sampled patches span the full morphological envelope of the five sites without clustering in any projection.

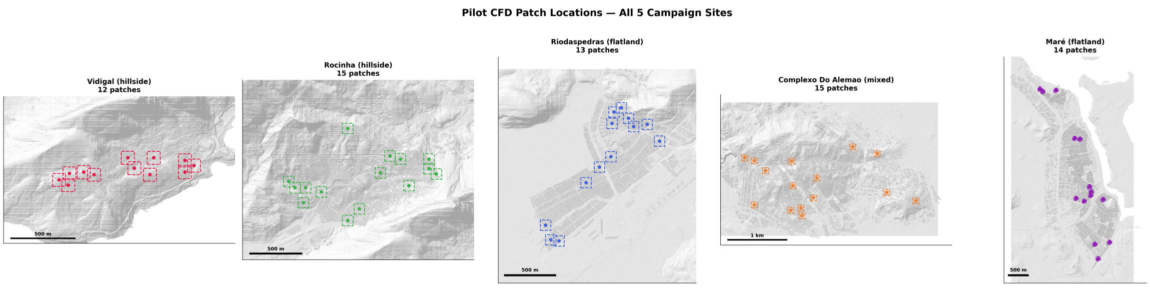

Figure (all sites overview). 5-panel map of final patch locations per site, each with hillshade basemap and patch outlines in site colour.

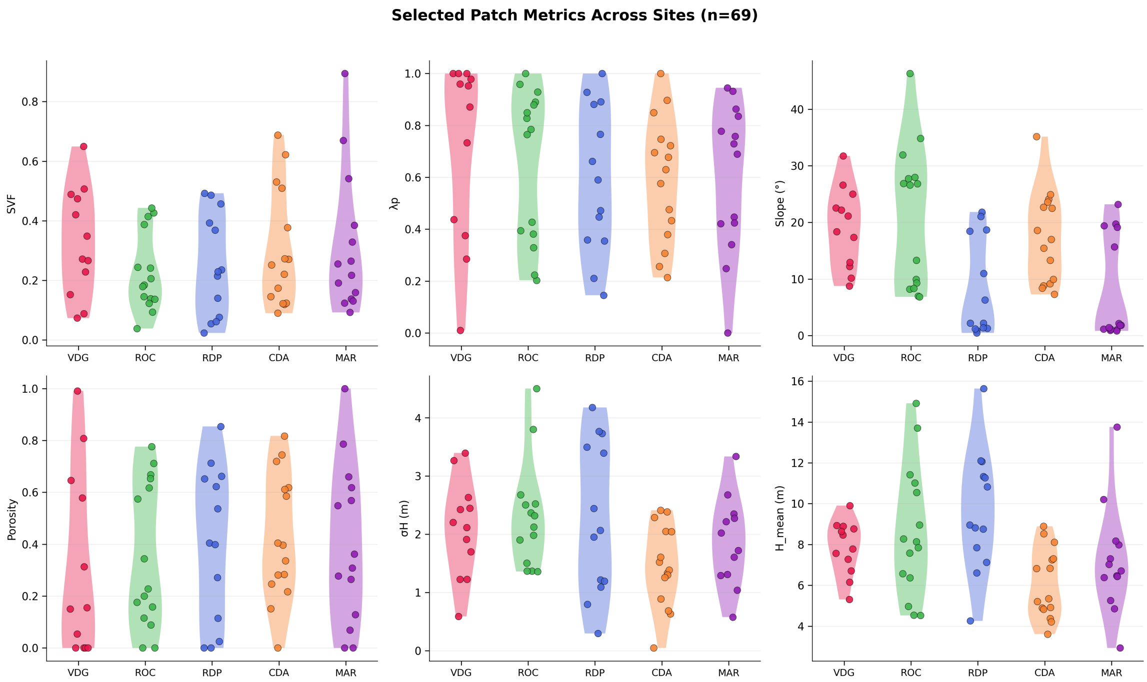

Figure (patch metrics). Violin plots of morphometric values at the 119 selected patches, grouped by site. Confirms that the selection does not over-represent or under-represent any single site's typical conditions: each site's patch distribution spans its full grid-cell distribution for each indicator.

Final per-site allocation:

Site	Pilot	Campaign add	Total	Shortfall
Vidigal	12	10	22	0
Rocinha	15	10	25	0
Rio das Pedras	13	9	22	−1
Complexo do Alemão	15	10	25	0
Maré	14	11	25	0
TOTAL	**69**	**50**	**119**	**−1**

Rio das Pedras has a one-patch shortfall against its 23-patch target: the SVF2_SLP2_LP2 stratum has only 7 eligible cells, all within 80 m of existing pilot patches, so no additional candidate could be placed without violating the minimum-spacing constraint. This is a genuine morphological limitation (the stratum is physically scarce) and was accepted rather than relaxing the spacing.

6.5 Blocken compliance

For each selected patch, the maximum building height within the 100 m analysis zone is recorded (`H_max_analysis`) and the Blocken (2015) minimum-fetch requirement is computed as `5 × H_max`. All 119 patches satisfy the constraint (`blocken_ok = true` in every `patch_meta.json`): the 250 m domain radius exceeds `5 × H_max` at every patch, with margins ranging from 114 m (RDP-P15, `H_max_analysis` = 27.3 m) to 215 m (median 180 m). 11 of 119 patches have margins below 150 m, all at Rio das Pedras (5/22) or Rocinha (6/25) — the two sites with the tallest favela structures (max `H_max_analysis` = 27.3 m at RDP, 22.3 m at ROC). Vidigal (max 19.3 m), Maré (18.1 m), and Complexo do Alemão (15.7 m) clear the 150 m margin at every patch.

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7. CFD Integration Pipeline

7.1 Overview

The code to ingest CFD results, aggregate them onto the 10 m morphometric grid, and combine directional simulations into an annualised ensemble is implemented in `src/cfd_integration/` (~1,100 lines across five modules: `schema.py`, `io.py`, `aggregate.py`, `metrics.py`, `weighting.py`) and covered by **63 passing unit tests**.

Input contract (documented fully in `src/cfd_integration/README.md`, provided to the CFD agent as the delivery specification):

```
data/{site}/cfd_results/{patch_id}/{wind_direction}/
  sample_points.csv       # required — 15k rows at z = 1.5 m, 2 m spacing
  summary.json            # required — simulation metadata
  field.vtu               # optional — full 3D field
```

Plus `data/{site}/wind_rose.json` per site (now built from measured INMET BDMEP / Iowa ASOS records; see §2.3 and Figure S5).

7.2 Module architecture

- **`schema.py`** — dataclasses for sample points, per-simulation metadata, campaign aggregates, wind roses; defines the 8 cardinal directions and runtime validators.
- **`io.py`** — CSV + JSON loading (primary path), optional VTU loading via `pyvista` (fallback), directory traversal for per-site campaigns.
- **`aggregate.py`** — spatial aggregation onto the morphometric grid. Primary function `aggregate_to_grid()` maps CFD samples onto individual 10 m cells whose centroid falls inside the 100 m-diameter circular analysis patch (the scientifically defensible zone per Blocken 2015). `aggregate_to_domain()` is a supplementary function that includes the full 250 m radius for robustness checks.
- **`metrics.py`** — health-relevant scalar quantities:
  - `velocity_magnitude()` — $\sqrt{U^2 + V^2 + W^2}$

  - `stagnation_fraction()` — fraction of samples with |U| < threshold (default 0.5 m/s, the commonly cited calm-air limit)
  - `turbulent_intensity()` — TI = √(2/3 · TKE) / U_ref
  - `ach()` — air change rate using the standard urban-climate formulation ACH = 3600 × ⟨|U|⟩ / L
  - `low_wind_percentile()` — worst-case stagnation (default p10)
  - `canyon_ventilation_efficiency()` — ⟨|U|⟩ / U_ref
- **`weighting.py`** — wind-rose-weighted combination of 8 directional simulations into an annualised metric. Supports three weighting modes: uniform (1/8 each), frequency-only (∝ f_dir), and frequency × speed (∝ f_dir · U_dir, which emphasises directions contributing more wind energy). `worst_case_direction()` finds the direction maximising any chosen metric (e.g. the direction producing most stagnant flow).

7.3 CFD agent specification

The full delivery contract (CSV schema, JSON schema, directory layout, methodology requirements for turbulence model, boundary conditions, meshing, post-processing, and convergence targets) is documented in **`src/cfd_integration/README.md`**. Key decisions baked into the contract:

- **Simulation domain:** cylindrical, 250 m radius, ≥ 6 × H_max height (≥ 90 m). Pedestrian sampling only valid within the central 100 m- diameter circular analysis patch.
- **Turbulence model:** k-ω SST (standard RANS for ABL flows at this scale).
- **Inlet:** logarithmic velocity profile, suburban roughness class (z_0 ≈ 0.03 m), turbulence intensity ≈ 0.15 at 10 m.
- **Simulations per site:** 8 cardinal directions, sharing one mesh with rotated inlet BC. 119 patches × 8 directions = 952 total simulations for the full campaign.
- **Sampling:** horizontal slice at z = 1.5 m, 2 m in-plane spacing, OpenFOAM `sample` utility, delivered as CSV.
- **Recommended validation patch: MAR-P07** (Maré, flat, 12.2 m max height, 1,200 buildings in domain) — the simplest end-to-end test case to validate the full pipeline before scaling.

7.4 End-to-end usage (post-CFD)

```python
from src.cfd_integration import (
    load_campaign_results, aggregate_to_grid, weighted_by_wind_rose,
)

campaign = load_campaign_results("vidigal")  # reads data/vidigal/cfd_results/
for patch_id in campaign.patch_ids():
    per_direction = {
        dir_: aggregate_to_patch(campaign.patches[patch_id][dir_],
                                 patch_center_xy)
        for dir_ in campaign.directions_for(patch_id)
    }
    annual = weighted_by_wind_rose(per_direction, campaign.wind_rose,
                                   weight_by="freq_speed")
    # annual = {annual_U_mean, annual_stagnation_frac, annual_ach, ...}
```

---

8. Repository Structure

Canonical output layout (as of 2026-04-13):

```
outputs/
├── {site}/
│   ├── morphometrics/         # SVF, per-building metrics, 10m grid,
│   │   ├── svf/              #   figures, report
│   │   ├── buildings/
│   │   ├── grid/
│   │   ├── figures/
│   │   └── report/
│   ├── morphometrics_20m/    # 20m grid (for resolution sensitivity)
│   └── sampling_cfd/
│       ├── pilot_sampling/   # 12–15 patches
│       └── campaign_sampling/ # 22–25 patches (pilot + new)
│
├── comparative/               # cross-site analysis
│   ├── pilot_summary/        # pilot-phase cross-site comparison
│   ├── final_allocation/     # 119-patch campaign summary
│   └── audits/               # topology audits
│
└── paper_figures/             # manuscript figures
    ├── fig_style.py          # shared style module
    ├── fig01_study_sites.py
    ├── fig02_morphometric_distributions.py
    ├── fig03_svf_lambda_coupling.py
    ├── fig04_sampling_design.py
    ├── fig05_morphometric_maps.py
    ├── figS1_correlation_matrices.py
    ├── figS2_context_extension.py
    ├── figS3_resolution_sensitivity.py
    ├── figS4_patch_thumbnails.py
    ├── figS5_wind_roses.py
    ├── README.md
    └── exports/              # rendered PNG + SVG
```

Source modules:

```
src/
├── config.py                  # paths, CRS, filtering thresholds
├── cartography.py             # scale bars, north arrows, UTM formatting
├── morphometry/               # grid computation
│   ├── grid.py
│   ├── indicators.py
│   ├── audit.py
│   ├── figures.py
│   └── report.py
├── svf_v2/                    # SVF engine (Tregenza-145 ray-cast)
│   ├── compute.py            # per-point and parallel raycast drivers
│   ├── scene.py              # 3D scene from DTM + footprints
│   ├── sampling.py           # passageway / street-centreline samplers
│   ├── paths.py              # per-site path registry
│   ├── io.py                 # per-cell aggregation + raster I/O
│   ├── utils.py
│   ├── facades.py            # façade-side sampling
│   └── visualize.py
├── solar/                     # solar access (phase 4)
└── cfd_integration/           # CFD ingestion (phase 7)
    ├── schema.py
    ├── io.py
    ├── aggregate.py
    ├── metrics.py
    ├── weighting.py
    └── README.md             # CFD agent specification
```

Key scripts:

Script	Purpose
`run_morphometric_audit.py`	Per-site morphometric grid + SVF audit + report
`build_extended_context.py`	Site + city-wide building and DTM merger
`run_pilot_sampling.py`	12-strata pilot batch (12–15 patches per site)
`run_campaign_sampling.py`	Campaign allocation (SVF-priority, target-total)
`build_wind_rose.py`	INMET → wind_rose.json ingestion

---

9. Validation Summary

Check	Status	Evidence
Building geometry validity	PASS (10 repaired, all sites clean)	`outputs/comparative/audits/rocinha_topology_audit.md`
CRS consistency	PASS	all inputs verified EPSG:31983 on load
SVF passageway aggregation (no centroid artefacts)	PASS	diagnostic in `src/morphometry/grid.py::_aggregate_svf_to_grid`
DTM sentinel value handling	PASS (post-fix)	sentinel = −9999 pixels masked in extended-context merge per `scripts/build_extended_context.py`
Extended context eligibility recovery	PASS	Vidigal eligible cells go from ~0.45 to 0.80 of grid with the 300 m extension; see `outputs/comparative/pilot_summary/candidate_pool_comparison.csv`
Blocken fetch compliance (all 119 patches)	PASS	`blocken_ok = true` in every `patch_meta.json`; margins 114–215 m, median 180 m, with 11/119 < 150 m (all at RDP or Rocinha)
Minimum inter-patch spacing (80 m)	PASS	realised minimum: 80–85 m across sites
Resolution sensitivity (10 m justified)	PASS	Figure S3; 10 m captures features 20 m loses
12-strata coverage (≥ 1 site per stratum)	PASS	all 12 strata non-empty when pooled
CFD integration unit tests	PASS	63 / 63 tests

---

10. Known Limitations

1. **Neutral stability is assumed.** All five `wind_rose.json` files now carry measured hourly observations (`quality_flag: "measured"`, 2015–2024 window, n = 64,088–89,439; see §2.3). The methodological simplification that remains is the assumption of neutral atmospheric stability across all directions and seasons. Stability classes are not separated; CFD inflow uses the neutral log-law profile. In practice this is conservative for the dispersion-relevant low-wind regimes that dominate Rio das Pedras and Complexo do Alemão (calm fractions 46 % and 33 %), and a known limitation for the more stably stratified nocturnal hours.

2. **CFD results not yet integrated.** `src/cfd_integration/` is tested but has not yet processed real simulation data. Any assumptions about sample-point density, column naming quirks, or edge cases in OpenFOAM output will only surface at first ingestion. The first pilot patch (VDG-P07) is in flight; running the test suite plus `cfd-results-ingestor` agent against the returned results will catch most issues.

3. **SVF validated against UMEP across all 5 sites** (limitation closed 2026-04-29 for Vidigal; height-matched at z = 1.5 m for Vidigal and Maré 2026-04-30; extended to all 5 sites 2026-05-01). The Tregenza 145-patch engine was cross-validated against UMEP's shadow-casting SVF (`svfForProcessing153`) at z = 1.5 m after lowering building heights by 1.5 m to height-match the two engines:

Site	n	r^2	slope	RMSE	bias
Vidigal	2,510	0.68	0.96	0.12	+0.01
Rocinha	5,646	0.81	1.01	0.14	+0.09
Rio das Pedras	1,503	0.93	1.25	0.11	+0.08
Complexo do Alemão	4,838	0.76	0.92	0.17	+0.14
Maré	9,516	0.94	0.97	0.12	+0.09

   Four of five sites agree at slope within ±8 % of unity (Vidigal, Rocinha, Complexo do Alemão, Maré). Rio das Pedras at slope = 1.25 reflects its small valid grid (n = 1,503) concentrated in the 0.3–0.5 SVF band where the 153-patch shadow-cast and 145-patch ray-cast integrations diverge most. Both engines are defensible operational definitions of SVF; the 5-site slope-≈-1 cluster establishes that the IVF Tregenza-145 ray-cast engine is consistent with an independent benchmark across the full morphological range represented in the campaign. See §4 cross-validation table for per-site interpretation and `outputs/{site}/morphometrics/svf/umep_validation/scatter.png` (S6 Vidigal, S7 Maré, S8 Rocinha, S9 Rio das Pedras, S10 Complexo do Alemão) for per-cell scatters.

4. **Resolution sensitivity is 10 m vs 20 m only.** Finer grids (5 m, 2 m) would be prohibitively expensive at site scale but warrant a spot-check for the CFD patches specifically. Not a priority for the current milestone.

5. **Cidade de Deus excluded.** A data integrity issue in the CDD building footprints (geometry inconsistencies; see project memory `project_cdd_data_bug`) leaves CDD outside the 5-site campaign. This is a final decision for the current campaign cycle: the 119-patch allocation, OpenFOAM submission, and downstream analysis all assume 5 sites. Re-onboarding CDD is out of scope until the building data is reprocessed upstream of this repository.

---

11. Next Steps

In this repository:

▣

Wind ingestion — measured roses for all 5 sites (completed 2026-04-27, see §2.3)

- [x] Build the result-side analysis pipeline (`scripts/analyze_cfd_results.py`) end-to-end on synthetic CFD data so the chain is exercised before VDG-P07 returns (completed 2026-04-29; smoke-tested across all 5 sites 2026-04-30: 359–405 covered cells per site, predictor signs consistent with morphology). Producer↔validator loop closed via `cfd-results-ingestor` against the synthetic outputs; both surfaced WARNs fixed in `scripts/generate_synthetic_cfd_results.py` so future synthetic runs hit PASS.
- [x] Cross-validate SVF against UMEP shadow-cast reference at pedestrian height across all 5 sites (closed 2026-05-01; slopes 0.92–1.25 with four of five within ±8 % of unity; see §10.3 and §4).
- [ ] Optional: 20 m grids already exist; generate the variant Fig S3 forms (`--variants` flag) if reviewers request the scatter or difference-map views
- [ ] Draft the Nature Cities manuscript (separate from this technical report) once the result-side pipeline is operational

In the CFD repository:

- [ ] Implement OpenFOAM case from per-patch exports (template in `outputs/{site}/sampling_cfd/campaign_sampling/patches/{id}/`)
- [ ] Validate pipeline on VDG-P07 (in flight at MIT ORCD)
- [ ] Mesh convergence study
- [ ] Run full 952-simulation campaign
- [ ] Post-process to `sample_points.csv` + `summary.json` per simulation

After CFD results land here. All of the machinery for these steps was built and synthetic-validated in commits `c9d25d6` (pipeline) and `0851b5e` (synthetic-generator self-validation); each box flips when the same code path is re-run against real Airflow output.

- [ ]

Verify first patch ingests correctly via `cfd-results-ingestor` (validator already accepts both IVF-native CSV and Airflow- native parquet layouts; tested against synthetic 160/160 cleanly)

- [ ] Annualise via wind-rose weighting (`src.cfd_integration.weighting`) — implemented; default weight is `freq_speed`
- [ ] Map CFD metrics onto the 10 m grid (nearest-patch v0) — implemented; covered cells 359–405 per site under synthetic
- [ ] Extend Figure 5 to include wind-velocity panel (`outputs/{site}/cfd_analysis/figures/fig5_wind_panel.png` generated under synthetic; awaits real numbers for the report)
- [ ] Statistical analysis: SVF/λp/terrain as predictors of ACH and stagnation — predictor regression CSV generated per-site under synthetic with expected sign structure
- [ ] Health-outcome linkage

Appendix A — Figure Index

#	Title	File
1	Study sites overview	`fig01_study_sites.png`
2	Morphometric distributions	`fig02_morphometric_distributions.png`
3	SVF–λp structural coupling	`fig03_svf_lambda_coupling.png`
4	CFD sampling design	`fig04_sampling_design.png`
5	Morphometric spatial maps	`fig05_morphometric_maps.png`
S1	Correlation matrices (5 sites)	`figS1_correlation_matrices.png`
S2	Extended context validation	`figS2_context_extension.png`
S3	Resolution sensitivity (10m vs 20m)	`figS3_resolution_sensitivity.png`
S5	Wind roses (5 sites, measured)	`figS5_wind_roses.png`
S6	UMEP cross-validation of SVF (Vidigal)	`figS6_umep_validation.png`
S7	UMEP cross-validation of SVF (Maré)	`figS7_umep_validation_mare.png`
S8	UMEP cross-validation of SVF (Rocinha)	`figS8_umep_validation_rocinha.png`
S9	UMEP cross-validation of SVF (Rio das Pedras)	`figS9_umep_validation_riodaspedras.png`
S10	UMEP cross-validation of SVF (Complexo do Alemão)	`figS10_umep_validation_complexo_do_alemao.png`
—	Strata heatmap	`fig_strata_heatmap.png`
—	Candidate pool breakdown	`fig_candidate_pool.png`
—	Campaign allocation summary	`fig_campaign_allocation_summary.png`
—	Cross-site feature space	`fig_campaign_cross_site_featurespace.png`
—	All sites patch overview	`fig_all_sites_patches.png`
—	Patch metrics comparison	`fig_patch_metrics_comparison.png`

All figures in `docs/technical_report/figures/`.

Appendix B — Commit History (Relevant to CFD Campaign)

```
4dac8e8 feat: CFD integration module + Fig S3 resolution sensitivity
1890123 docs: update README and ROADMAP for v5.0.0 — CFD campaign complete
43f8554 feat: Nature Cities paper figures — 9 figures + shared style module
571ba47 feat: CFD sampling pipeline — stratified patch selection + campaign allocation
5e6e4c1 docs(figS3): canonicalize distribution-overlay as the main figure
9768e94 feat(wind-rose): scaffolding + template generator for CFD wind forcing
d477935 test(cfd-integration): 46 tests covering schema, IO, aggregation, metrics, weighting
```

Appendix C — Key References

- Blocken, B. (2015). Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations. *Building and Environment*, 91, 219–245.
- COST Action 732 (2007). Best practice guideline for the CFD simulation of flows in the urban environment.

- Stewart, I. D. & Oke, T. R. (2012). Local Climate Zones for Urban Temperature Studies. *Bulletin of the American Meteorological Society*, 93(12), 1879–1900.
- Tregenza, P. R. (1987). Subdivision of the sky hemisphere for luminance measurements. *Lighting Research & Technology*, 19(1), 13–14.

Prepared in `docs/technical_report/`. Available in both markdown (`technical_report.md`) and PDF (`technical_report.pdf`) formats.

Regenerate the PDF:

```
python docs/technical_report/build_pdf.py
```

Pipeline: pandoc (markdown → HTML) then weasyprint (HTML → PDF). Requires `pandoc` and the `weasyprint` Python package.

Regenerate all figures:

```
for f in outputs/paper_figures/fig*.py; do python3 "$f"; done
```

Regenerate campaign allocation:

```
python scripts/run_campaign_sampling.py
```